

TIM TOMASHEVSKIY

AI Safety Researcher • Reinforcement Learning & Sequential Decision-Making • Runtime Assurance for Autonomous Systems

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RESEARCH PROFILE

PhD candidate at McMaster University researching reinforcement learning and safety for autonomous systems operating under nonstationarity. Work spans safe and continual reinforcement learning, adaptive safety constraints, proactive context forecasting, runtime assurance, and evaluation under distribution shift — with the long-term goal of making adaptation itself a subject of safety assurance. Earlier work in automotive functional safety grounds this research in a systems-safety approach to learning-enabled and autonomous systems.

Core expertise: Safe RL • Continual / Adaptive Learning • Nonstationarity & Distribution Shift • Runtime Assurance & Shielding • Sequential Decision-Making • Autonomous Systems • Functional Safety

RESEARCH EXPERIENCE

PhD Researcher — AI Safety & Reinforcement Learning

McMaster University, Department of Computing and Software

2022 – Present

Hamilton, ON

- Conduct an independent research program on maintaining safety as environmental conditions, learned knowledge, and safety constraints evolve over time.
- Designed context-dependent and adaptation-speed safety constraints for nonstationary RL, including mechanisms for detecting when environmental change exceeds an agent's safe adaptation capacity.
- Developed proactive, context-forecasted safety constraints that infer latent context, forecast its evolution, quantify predictive uncertainty, and construct anticipatory constraints before unsafe behavior emerges.
- Built CPSS (Constraint Projection Safety Shield), a runtime mechanism converting cumulative safety budgets into enforceable, action-level constraints; **reduced average collision rate by 91.6% (0.082 → 0.007) versus an unshielded DQN baseline across four highway-env driving tasks.**
- Architected LILAC+, a modular framework integrating context-based, adaptation-speed, and soft-to-hard safety constraints; reduced safety violations relative to unconstrained and fixed-constraint baselines across stationary, seen-nonstationary, and unseen-nonstationary regimes while maintaining competitive reward.
- Designed and ran RL experiments (DQN/PPO) in nonstationary driving environments; **CoLLAs 2026 evaluation achieved an 87.2% collision-rate reduction on the main benchmark and 92–97% reduction across three held-out road layouts.**
- Developing SafeShiftRL, a modular benchmark for evaluating safety under stationary, seen-nonstationary, and unseen-nonstationary conditions — covering constraint validity, physical safety, intervention burden, and adaptation behavior.

Research Assistant / Research Software Developer — Automotive Functional Safety 2019 – 2022

McMaster Centre for Software Certification (McSCert) — LEAP/AVIN project with Stellantis Hamilton, ON

- Contributed to research on functional safety of centralized automotive electrical/electronic (E/E) architectures and domain controllers.
- Developed and validated models and software for electric-vehicle powertrain and communication systems using MATLAB, Python, C/C++, TensorFlow, and PyTorch.
- Investigated safety architectures for catastrophic loss of a centralized domain controller; contributed to a cost-efficient fallback architecture preserving essential vehicle functionality, feeding into the project's system-level safety case for Stellantis.

- Co-authored an IEEE/IFIP DSN-W 2021 paper; presented research in Stellantis project meetings and at the 7th International Workshop on Safety and Security of Intelligent Vehicles (SSIV 2021).

SELECTED PUBLICATIONS

- [Proactive Context-Forecasted Safety Constraints for Nonstationary Reinforcement Learning](#) — CoLLAs 2026, accepted; arXiv:2609.08080; sole author
- [Safe Continual Reinforcement Learning under Nonstationarity via Adaptive Safety Constraints](#) — arXiv preprint, 2026; arXiv:2605.18842; sole author
- [From Cumulative Constraints to Adaptive Runtime Safety Control for Nonstationary Reinforcement Learning](#) — arXiv preprint, 2026; arXiv:2605.18841; sole author
- [Adjustment Speed as a Safety Constraint for Nonstationary Reinforcement Learning](#) — arXiv preprint, 2026; arXiv:2607.21646; sole author
- [Safe Continual Reinforcement Learning Methods for Nonstationary Environments: Towards a Survey of the State of the Art](#) — Survey; arXiv preprint, 2026; arXiv:2601.05152; sole author
- [A Safety Architecture for Centralized E/E Architectures](#) — IEEE/IFIP DSN Workshops, 2021; with V. Bandur, V. Pantelic, M. Lawford

PRIOR EXPERIENCE

Data Analyst / Architect / Lead Developer

CGStar Technology

2013 – 2019

Toronto, ON

- Data Analyst (2017–2019): built SQL/Tableau reporting infrastructure and designed product experiments that surfaced process and business improvements; led a small team delivering data products for internal decision-making.
- Architect / Lead Developer (2013–2017): designed Java/J2EE architectures for complex software systems and translated mathematical and statistical models into production implementations, including algorithms for discrete-event and machine-learning systems.

EDUCATION

PhD Candidate, Computing and Software — McMaster University (Expected 2026)

MBA — Weatherhead School of Management, Case Western Reserve University

M.Sc., Applied Mathematics and Information Technology — Moscow State Institute of Radio Engineering, Electronics and Automation

B.S. (Honours), Computer Science — MIREA, Faculty of Automation

TECHNICAL SKILLS

Machine Learning & Research: Reinforcement Learning (DQN, PPO), Safe RL, Constrained RL, Continual Learning, Sequential Decision-Making, Deep Learning, Uncertainty-Aware Forecasting, Experimental Design

AI Safety: Adaptive Safety Constraints, Runtime Assurance / Shielding, Safety under Nonstationarity, Distribution Shift, Functional Safety, Safety Evaluation

Programming & Tools: Python, PyTorch, TensorFlow, MATLAB, C/C++, Java/J2EE, SQL, Tableau